

MOHAMED AL JALANJI

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France



PROFESSIONAL SUMMARY

Data science & AI practitioner with a hybrid data science and computer-science background, focused on LLMs, agentic systems, LLMOps and MLOps. Experienced in end-to-end agentic and model development, fine-tuning, deployment and evaluation across self-hosted and AWS environments, and consistently engaged with the latest AI research and techniques.

WORK EXPERIENCE

Lead AI Architect

Vaxx Research

June. 2026 – Now

Calgary (Remote), Canada

- Leading the design and implementation of an AWS-native agentic AI platform for autonomous mineral-process execution, integrating Bedrock models, SCADA data flows, and Aspen HYSYS simulations.
- Defining the safety, governance, and validation framework for production deployment, including Z3 guardrails, human oversight, and auditable operational controls.

Data Scientist & AI Engineer (LLMs)

PrediSurge

Nov. 2025 – June 2026

Saint-Etienne, France

- Led the end-to-end architecture and implementation of an agentic RAG pipeline on structured clinical data and PDFs using fine-tuned SLMs, PydanticAI-based orchestration, AWS Bedrock, and LLMOps (prompt versioning, monitoring, RAGAS evaluation).
- Designed a scalable, secure, and low-latency integration architecture for the agentic RAG within the company's SaaS platform built on AWS microservices.

AI Engineer

APTIV

Oct. 2023 – Sep. 2025

Vannes, France

- Transformed optimally dataflow graphs to Logical Execution Time (LET) design for automobile projects using Constraint Programming (Z3), ensuring the reduction of LET design manual labor by almost **70%**.
- Fine-tuned the LLM model CodeT5 using PyTorch to translate text to a local automobile platform command language, and developed evaluation metrics & parsers needed, achieving an accuracy of **91%**.
- Developed Cloner tool for systematic duplication of AUTOSAR Classic elements using lxml Python library to reduce time-execution errors for a local SensorFusion project running on the target automobile platform.

ML Engineer Intern

ETIS lab (CNRS UMR 8051)

May 2022 – Jul. 2022

Cergy, France

- Applied the MPDist distance measure using the Stumpy library to construct distance matrices on linguistic burst time series data, achieving a DBCV clustering validation score of **0.73** using HDBSCAN.

Junior Data Scientist

PI Marketing

Mar. 2017 – Sep. 2019

Tangier, Morocco

- Deployed ML ensemble models for customer segmentation and predictive lead scoring to optimize marketing campaign performance and personalize outreach using Pandas, Scikit-learn, SQL & Tableau.

Android Developer

Smart Source

Feb. 2016 – Jan. 2017

Tangier, Morocco

- Developed an Android app enhancing operational efficiency for clients using Android Studio & Postman.

EDUCATION

Data ScienceTech Institute <i>Master in Data Science & Artificial Intelligence</i> GPA : 4.0	Oct. 2023 – Dec. 2025 <i>Paris, France</i>
Université Sorbonne Paris Nord <i>Bachelor in Computer Science</i> GPA : 3.7	Sep. 2021 – Jul. 2023 <i>Villetaneuse, France</i>
Seconda Università di Napoli <i>Bachelor in Statistics & Data Analytics</i> GPA : 3.7	Sep. 2020 – Oct. 2023 <i>Caserta, Italy</i>

PROJECTS

<https://aljalanjimohamed.dev/index.html#projects>

CERTIFICATIONS

- AWS Certified Solutions Architect
- Deep Learning Specialization — DeepLearning.AI
- Neo4j
- Discrete Optimization — Georgia Tech

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Cypher, C/C++, Kotlin, NetLogo

Data & ML & DL: Pandas, NumPy, PySpark, Scikit-Learn, PyTorch, UnslothAI, DistillKit (Distillation)

LLMOps & Agentic AI: LangChain, HayStack, PydanticAI, Strands, DSPy, vLLM, NiceGUI, RAGAS, DeepTruth, Langfuse

MLOps & DevOps: MLflow, DVC, Docker, GitHub CI/CD, FastAPI

Cloud: AWS (Bedrock, SageMaker, Inferentia, Athena, Lake Formation, S3, RDS, EC2, ECS, EKS)

IaC: Terraform

Databases: PostgreSQL, Neo4j

Tools: Git, VS Code, Claude Code

INTERESTS & LANGUAGES

Languages: English (native), Arabic (native), French (advanced), Italian (beginner)

Interests: Blues & Bluegrass & Country Music, Philosophy Readings, Canoeing, Hiking, Mountain Biking